

Internal Knowledge Assistant

Complete Technical Architecture & End-to-End Execution Flow

V1.0 · ZERO-SQL LOCAL AI

1. Architecture Overview

The **Internal Knowledge Assistant** is an enterprise-grade, privacy-first Retrieval-Augmented Generation (RAG) system. It combines a **Zero-SQL in-memory data store**, **SIMD-accelerated FAISS vector search**, and local **Ollama AI models** with strict pre-retrieval Access Control Lists (ACLs).

⚡ In-Memory + JSON Store

Zero database dependency.
Document metadata and chunks are held in thread-safe memory and persisted directly to `chunks.json`.

🎯 Dense FAISS Search

`IndexFlatIP` inner-product search with cosine normalization & SIMD vector acceleration using 384-dim BAAI embeddings.

🖨️ Local LLM (Ollama)

Powered by `qwen2.5:1.5b`
configured for a lean 256-token context window with fine-grained chunk retrieval (<80 tokens).

2. End-to-End Query & Generation Flow

Every query sent to `POST /api/chat` passes through a deterministic 6-stage lifecycle orchestrated by `RAGService.cs`:

STAGE 1 Pre-Retrieval ACL Authorization Filter

Examines user Persona claims (Roles, Department, AccessLevel 1–5). Computes the strict set of permitted `documentIds` before vector search starts. Unpermitted files are completely invisible.

Service: `AuthorizationService.cs` | Avg Latency: 2–4 ms

STAGE 2 Dense Vector Embedding Generation

Sends the user query text to Ollama (`/api/embed`) to produce a 384-dimensional dense semantic embedding vector using `BAAI/bge-small-en-v1.5`.

Service: `OllamaEmbeddingService.cs` | Model: `bge-small-en-v1.5` (384d) | Avg Latency: 20–35 ms

STAGE 3 FAISS IndexFlatIP Vector Search & Top-K Ranking

Executes SIMD dot-product cosine similarity search against all permitted chunk embeddings. Applies a minimum relevance score cutoff (`minScore = 0.35`) and selects Top-2 candidate chunks.

Service: `FaissVectorStore.cs` | Metric: Dot Product (IndexFlatIP) | Avg Latency: 10–30 ms

STAGE 4 Clearance & Query Policy Evaluation

Applies safety guardrails (e.g. personal salary/compensation queries, unreleased board projects). If restricted or if 0 chunks pass clearance, immediately refuses with: *"You do not have access to this information."* without leaking source citations.

Service: `RAGService.IsRestrictedQuestion` | Avg Latency: 1 ms

STAGE 5 Grounded LLM Generation (Qwen 2.5 1.5B)

Constructs a compact grounded prompt incorporating the authorized chunks. The local LLM generates a concise, accurate answer with mandatory page citations (`[Doc: Title, Page X]`).

Service: `OllamaLLMService.cs` | Context Window: 256 tokens | Avg Latency: 5–8 sec

STAGE 6 Citation Packaging & Audit Trail Logging

Builds citation links with page targets (`/api/documents/{id}/file#page=X`), measures execution time for each stage, and writes an immutable audit record in memory. Returns the final

`ChatResponse` DTO.

Service: `AuditService.cs` | Output: `ChatResponse` JSON | Avg Latency: 2–5 ms

✦ 3. Chunking Architecture (256-Token Context Budget)

To achieve high inference speed and prevent context truncation on 256-token context windows, documents are partitioned into sentence-level granular chunks:

Fine-Grained Chunking Parameters

- **Target Token Size:** 65 tokens (~260 characters)
- **Maximum Token Cap:** 85–92 tokens
- **Overlap Window:** 12 tokens
- **Split Boundaries:** Sentences, bullet points (•, ✓, ❖, ➤), newlines
- **Total Granular Chunks:** 402 chunks across 8 documents

256-Token Prompt Budget Breakdown

- **System Prompt:** ~35 tokens (Identity & strict grounding rules)
- **User Query:** ~15–25 tokens
- **Top-2 Retrieved Chunks:** ~90–120 tokens (45–60 tokens each)
- **Prompt Total:** ~140–180 tokens
- **Available Output Space:** ~76–116 tokens for generated answer

🛡️ 4. Security Clearance & Persona Access Matrix

Pre-retrieval access enforcement guarantees multi-tenant role isolation at every layer:

Persona	User ID	Department	Level	Accessible Knowledge Base
David Chen	usr-david-01	General	Level 1	General Employment, Leaves, Code of Ethics, Sexual Harassment, Professional Conduct
Sarah Jenkins	usr-sarah-02	HR	Level 2	All Level 1 + HR Disciplinary Procedures, Grievance Filings, Performance Appraisals
Alex Mercer	usr-alex-03	Engineering	Level 3	All Level 1–2 + Architecture Blueprints, API specs, Cloud Infrastructure
Marcus Vance	usr-marcus-04	Finance	Level 4	All Level 1–3 + EBITDA reports, CapEx budgets, Financial Projections
Elena Rostova	usr-elena-05	Executive	Level 5	Full Enterprise Access (M&A briefings, Executive compensation, Board Minutes)

🚩 5. Core API Request & Response Specification

POST /api/chat Request

```
{
  "question": "What are the leave rules for employees on probation?",
  "userClaims": {
    "userId": "usr-david-01",
    "userName": "David Chen",
    "roles": ["Employee"],
    "department": "General",
    "accessLevel": 1
  },
  "topk": 2
}
```

POST /api/chat Response

```
{
  "messageId": "60a01a26-985b-4c88-8236-430cb16a6f14",
  "question": "What are the leave rules for employees on probation?",
  "answer": "According to the \"Employment Policies\" document (p.5), employees on probation are not eligible",
  "sources": [
    {
      "documentTitle": "Leave Policy",
      "pageNumber": 1,
      "relevanceScore": 0.831,
      "fileUrl": "/api/documents/c9f9.../file#page=1"
    }
  ],
  "securityInfo": {
    "evaluatedUser": "David Chen",
    "evaluatedAccessLevel": 1,
    "enforcementMode": "Pre-Retrieval Strict ACL Filtering"
  },
  "executionTimeMs": 6406,
  "stageTimingsMs": {
    "aclFilterMs": 3,
    "embeddingMs": 28,
    "vectorSearchMs": 29,
    "securityEvaluationMs": 1,
    "llmGenerationMs": 6341,
    "postProcessingMs": 6
  }
}
```